RESEARCH ARTICLE

Evaluation of Swallowing in Poststroke Individuals Using the Swallowing Proficiency for Eating and Drinking (SPEAD) Protocol

A Pilot Study

Uzair Chilwan,[1] Aswathy Suresh,[1] Sahlah Shameer,[1] Anaan Zohara,[2] and Radish Kumar Balasubramanium[1]

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Correspondence
Dr. Balasubramanium
radheesh.slp@manipal.edu

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Abstract

Background and Objectives

Poststroke dysphagia affects 37–78% of stroke survivors, leading to complications such as malnutrition, dehydration, and psychological distress. Current tools for assessing swallowing, such as the Timed Water Swallow Test and Test of Mastication and Swallowing Solids, do not evaluate intermediate consistencies. The Swallowing Proficiency for Eating and Drinking (SPEAD) protocol incorporates thin liquids, solids, and moderately thick liquids, providing a comprehensive evaluation of swallowing function. This study assessed SPEAD's utility in a stroke population and its relationship with stroke severity.

Methods

A cross-sectional study recruited 77 poststroke participants (aged ≥18 years). Swallowing was evaluated using SPEAD for thin liquids, moderately thick liquids, and solids, following International Dysphagia Diet Standardization Initiative guidelines. Parameters recorded included number of swallows, chews, grams consumed, and duration (in seconds), with derived measures such as ingestion speed and SPEAD rate. Stroke severity was assessed using the NIH Stroke Scale, and self-reported swallowing difficulties were screened with Eating assessment Tool (EAT-10). Data were analyzed using analysis of variance (ANOVA), multivariate analysis of variance, and Pearson correlation.

Results

Participants with moderately severe strokes demonstrated slower ingestion speeds and increased effort across all consistencies. Significant differences were observed in SPEAD parameters between consistencies, with solids requiring higher ingestion times and effort. Stroke severity correlated moderately with SPEAD measures ($r = -0.45$ to -0.62, $p < 0.05$). A significant correlation was observed between SPEAD and EAT-10 scores, indicating that participants with higher dysphagia burden as perceived through SPEAD also reported greater perceived difficulty with swallowing on EAT-10.

Discussion

The SPEAD protocol effectively identifies swallowing impairments in stroke patients, particularly for intermediate consistencies. Its comprehensive approach provides valuable insights for dysphagia management, emphasizing the need for further validation in diverse populations. The study also highlights the value of objective bedside assessments such as SPEAD in conjunction with patient reported outcomes such as EAT-10. Together, they offer a comprehensive understanding of swallowing function poststroke and can guide early intervention strategies.

Introduction

Swallowing is a complex process that relies on the coordination of various muscle groups to ensure the smooth transit of food or liquid reducing the residual matter while protecting the airway.[1] As swallowing is controlled by cortical, subcortical and brainstem structures,

[1]Department of Audiology and Speech Language Pathology, Kasturba Medical College Mangalore, Manipal Academy of Higher Education, Manipal, India; [2]Department of Physical Medicine and Rehabilitation, First Neuro Hospital, Padil, Mangalore, Karnataka, India.

dysphagia is commonly seen during the acute phase of stroke. Research also indicates that up to 65% of acute stroke patients experience dysphagia.[2-4] Thus, poststroke dysphagia as a common clinical problem affects 37%–78% of stroke survivors,[5] posing a significant risk for serious complications such as malnutrition and dehydration.[3] Furthermore, the literature has reported 48% prevalence of dysphagia and associated pneumonia.[6] As a result, increased hospital readmissions and elevated mortality rates are commonly encountered in stroke survivors.[7,8] Moreover, dysphagia detrimentally impacts the swallowing-related quality of life by impeding the ability to share meals with loved ones causing severe psychological distress in individuals who were previously able to eat normally.[9,10]

Speech Language Pathologists (SLPs) conduct clinical swallow evaluation with different food consistencies for a comprehensive evaluation of swallowing. Administering multiple swallowing trials with both solid and liquid boluses allow SLPs to infer the functional integrity of the oral and pharyngeal phases of swallowing.[11] Standardized quantitative tests, such as the Timed Water Swallow Test (TWST)[12] and Test Of Mastication And Swallowing Solids (TOMASS),[13] are most commonly used to assess swallowing function in liquids and solids, respectively. TWST and TOMASS uses level 0 (thin liquids) and level 7 (regular) bolus consistencies, respectively, as per the International Dysphagia Diet Standardization Initiative (IDDSI).[14] These tests are grounded in evidence that swallowing mechanics differ between solid and liquid consistencies.[15] Research from the previous years have recommended using more than 1 consistency to evaluate dysphagia, as it provides more robust details and accurate description on swallowing skills.[11]

Swallowing Proficiency for Eating and Drinking (SPEAD) is a comprehensive assessment protocol designed to evaluate swallowing function across various consistencies, enhancing the diagnostic scope for dysphagia. It incorporates key elements from the TWST for thin liquids, TOMASS for solids, and a third component—moderately thick liquids (IDDSI level 3)—to provide a detailed analysis of swallowing mechanics across different textures.[16] The inclusion of moderately thick liquids addresses a crucial gap in dysphagia assessment, as intermediate consistencies often challenge swallowing efficiency due to their specific rheological properties. This enables clinicians to detect subtle impairments that might not manifest during thin or solid bolus trials. The SPEAD test evaluates multiple parameters categorized into raw and derived measures. Raw parameters include the number of swallows, number of chews, grams consumed, and the total duration required to ingest a specific consistency. Derived parameters consist of the speed of ingestion, average volume of swallow, and the SPEAD rate. The speed of ingestion is calculated as the ratio of grams consumed to the total duration of ingestion. The average volume of swallow is determined by dividing the grams consumed by the number of swallows. Finally, the SPEAD rate is computed as the mean speed of ingestion across the 3 tested consistencies. By integrating quantitative measurements with qualitative observations, the SPEAD framework can enable SLPs to develop tailored intervention plans for individuals who have swallowing difficulty poststroke. Karsten et al.[16] and Dodderi et al.[17] validated the SPEAD protocol, demonstrating its reliability in capturing the nuances of oral and pharyngeal phase coordination in head and neck cancer and healthy older adults, respectively. However, we do not have SPEAD performance data in stroke population. Given the lack of prior research in this domain, this study was conceptualized as a pilot effort to evaluate the utility of SPEAD protocol and to establish preliminary trends across stroke severity.

Methods

Study Design

The study employed a cross-sectional design to address the research objectives, with the principles outlined in the Declaration of Helsinki.

Participants

A total of 77 healthy individuals, aged 18 years and above (mean age 58.4 ± 13.8), were recruited for participation in this study. The sample size was calculated using the formula $n = Z^2\sigma^2/d^2$, where $Z = 1.96$ at 5% level of significance, $\sigma = 6.66$ SD[14] and $d = 1.5$ clinical significant difference. Participants included in the study were individuals who had experienced an ischemic or hemorrhagic stroke, with assessment conducted at least 7 days poststroke onset. We also included subacute stroke patients (up to 4 weeks postonset) who were clinically stable and met the inclusion criteria. Participants were made to undergo standardized bedside swallowing clinical examination to check if they were fit for oral intake. Participants failing in the bedside examination and relying entirely on nonoral feeding methods (nasogastric or percutaneous endoscopic gastrostomy tubes) were not enrolled as this precluded reliable evaluation of oral intake or swallowing function. Those with significant cognitive or comprehension impairments, profound expressive deficits, or anarthria that hindered effective communication or participation in assessments were also excluded. In addition, individuals with active medical complications, such as recurrent strokes, unstable cardiac conditions, or infections, were excluded to ensure the study population consisted of stable participants.

Materials

The administration of the SPEAD test involves a detailed assessment of swallowing performance in stroke patients. The 3 feeds were categorized based on IDDSI guidelines namely levels 0, 3, and 7. Participants were instructed to swallow 100 g of water (Level 0), 100 g of thickened milky mist curd (Level 3), and 6.67 g of Parle Monaco (Level 7) representing 3 consistencies which is in accordance with the SPEAD protocol. Eating assessment Tool (EAT-10) was used to identify participants who had risk of oropharyngeal or esophageal dysphagia. It has 10 statements which covers daily experiences of swallowing difficulties on a 5-point Likert scale with 0 indicating no swallowing problem and 4 indicating severe problem. The total score on the EAT-10 scale ranges from 0 to 40, with a score exceeding 3 suggesting the presence of self-reported swallowing difficulties.[18,19] The NIH Stroke Scale (NIHSS) was used to assess stroke severity. The NIHSS comprises 15 items evaluating domains such as speech, cognition, and motor control. Stroke severity is categorized as follows: 0 indicating no stroke, 1–4 representing a minor stroke, 5–15 indicating a moderate stroke, 16–20 reflecting a moderately severe stroke, and 21–42 signifying a severe stroke.[20] Participants falling under severe stroke category on NIHSS scale were excluded from the study as they commonly met the exclusion criteria, making them ineligible for the study.

Procedure

During the administration of SPEAD protocol, all the participants were seated comfortably in an upright posture in a chair either independently or with minimal assistance. Patients who could not be positioned in an upright sitting posture, even with support were excluded from the study, as this position was necessary to ensure safe and standardized administration of SPEAD protocol. However, inability to maintain an upright sitting posture was not treated as an independent exclusion category, as it overlapped with other exclusion criteria and was accounted for within the broader screening exclusions as illustrated in eFigure 2 (see supplementary material).

Before presentation of the 3 feeds such as levels 0, 3, and 7 based on IDDSI guidelines, participants were provided 10 mL of room temperature water to ensure adequate hydration of the oral cavity. The SPEAD testing followed standard instructions: participants were asked to "Eat (or drink) this comfortably in your usual style and say your name aloud after swallowing the entire portion." They were also instructed to avoid talking between swallows to ensure they fully consumed the feeds.

After completing the task, the researcher visually inspected the participants' oral cavity for any signs of residue in the oral vestibule. In addition, other indicators of dysphagia, such as oral spillage, throat clearing, coughing, choking, and nasal regurgitation, were monitored as well. To minimize potential consistency effects, the sequence for administering the 3 different feeds in the SPEAD protocol was randomized across the study. This approach aimed to eliminate any bias associated with the order of test feed presentation.

Each of these swallowing tasks were carefully monitored and video recorded for subsequent data analysis to ensure precise measurement of any swallowing abnormalities, such as delays, aspiration, or inefficient bolus transport. This video data allowed for objective analysis by clinicians and researchers, offering a more nuanced understanding of the mechanical and physiologic impairments in the swallowing process.

In addition to SPEAD test, EAT-10 was provided to screen for participants who are at risk for any swallowing difficulties based on their self-perception. The task for the participants was to read the 10 statements related to swallowing ability in their daily life and score them on a 5-point Likert scale (0–4) with a total score ranging from 0 to 40 with higher scores depicting greater self-perception of swallowing problem. To further explore the link between stroke severity and swallowing difficulties, the NIHSS was administered to evaluate the severity of neurologic deficits in the stroke population. This multifaceted approach provided a comprehensive understanding of the relationship between stroke severity and swallowing difficulties in this cohort.

Data Analysis

The principal investigator reviewed all the video recordings to ensure that it was captured based on guidelines from the SPEAD protocol for all the participants. To minimize potential rater bias, video recordings were anonymized by removing age and feed level labels (levels 0 and 3). Two SLPs, trained in evaluating the SPEAD protocol, independently analyzed the videos, extracting the following raw data for each of the 3 consistencies: (1) Total duration (seconds), (2) Grams swallowed, (3) Number of swallows, and (4) Number of chews, as specified originally in the SPEAD protocol.[16] These raw data were subsequently used to calculate the following performance metrics: (1) Ingestion speed (grams/s), (2) Average volume of swallow (grams/swallow), and (3) SPEAD rate (grams/s). eFigure 1 (see supplementary material) summarizes the raw and derived parameters covered under SPEAD protocol.

Statistical Analysis

Jamovi version 2.6.11 was used to perform statistical analyses. Descriptive statistics, including mean, SD, minimum, and maximum values, were calculated to assess overall participant performance. Inter-rater reliability between 2 raters was evaluated using the intraclass correlation coefficient (ICC). A 1-way analysis of variance (ANOVA) was conducted to analyze the effect of stroke severity on the SPEAD rate, with a significance threshold of 0.05. In addition, a multivariate analysis of variance was used to assess the impact of both swallowing severity and stroke severity on the number of bites and chews, with the significance level set at 0.05. Post hoc pairwise comparisons were performed using the Tukey test, maintaining a significance level of 0.05. A 1-sample t test was conducted to compare the SPEAD rate in the stroke population against normative data for the Indian population[17] across young healthy adults, middle-aged healthy adults, and older healthy adults. Pearson correlation analysis was used to estimate the relationship between stroke severity and performance on swallowing assessments.

Standard Protocol Approvals, Registrations, and Patient Consents

The study was conducted in compliance with the Declaration of Helsinki and approved by the Institutional Ethics Committee of Kasturba Medical College Mangalore. Written informed consent was obtained from all participants.

Data Availability

Anonymized data not published within this article will be made available by request from qualified investigators.

Results

A total of 77 individuals participated, comprising 26 women and 51 men. The severity of dysarthria and aphasia was recorded based on the respective item scores from the NIHSS. The distribution of participants based on severity of dysarthria and aphasia, site of lesion, and type of stroke is illustrated in eFigure 2 (see supplementary material). The administration of SPEAD test in poststroke individuals took 20–25 minutes to complete all the consistencies. The results of the various parameters of SPEAD test are presented in Table 1.

Pairwise comparison was performed to check for differences in the parameters across different severity of stroke. Table 1 summarizes the observed differences in duration of swallowing activity across stroke severity levels. The number of chews, assessed only for regular solids (Level 7), demonstrated an increasing trend with greater stroke severity; however, this difference was not statistically significant. Total duration increased with increasing stroke severity for thin liquids (Level 0) and moderately thick liquids (Level 3), with these differences reaching statistical significance. Although total duration also increased for regular solids (Level 7), this change did not reach statistical significance. Furthermore, no change was observed in number of swallows irrespective of stroke severity and type of consistency.

eFigures 3–5 (see supplementary material) depict the mean and SD of performance metrics of SPEAD test across the 3 consistencies among the 3 stroke groups. eFigure 3 shows a decrease in speed of ingestion as the consistency of the test item increased. In addition, a progressive reduction in speed of ingestion was observed with the increasing severity of stroke. Participants with minor stroke comparatively had higher speed of ingestion than participants with moderate and moderately severe strokes. ANOVA revealed significant difference ($p < 0.001$) for a speed of ingestion across severity of strokes.

eFigure 4 illustrates the variation in average volume of swallow across different consistencies and stroke severities. Notable variation was observed for level 0, while levels 3 and 7 exhibited mixed patterns. Average volume of swallow was found to reduce as severity of stroke increased. However, average volume of swallow was found to be more in moderately severe strokes as compared with minor and moderate category. The results from the ANOVA test showed that no significant differences were observed on average volume of swallow on varying severity of stroke.

eFigure 5 depicts an inverse relationship between SPEAD rate and severity of stroke amongst the participants. Participants with minor stroke were found to have higher SPEAD rate as compared with participants with moderate and moderately severe stroke. ANOVA results depicted a statistically significant difference existed between stroke severity.

Inter-Rater Reliability

Table 2 summarizes the ICC values for the SPEAD parameters. All parameters demonstrated ICC values greater than 0.80 ($p < 0.001$), indicating a high level of agreement in the SPEAD analysis performed by the 2 reviewers.

Test-Retest Reliability

Table 2 presents a detailed analysis of Cronbach alpha values for the SPEAD parameters.

Notably, all parameters demonstrated excellent internal consistency, with Cronbach alpha values exceeding 0.9. However, the average volume of swallow across all the 3 consistencies exhibited a Cronbach alpha ranging from 0.84 to 0.86, reflecting good internal consistency.

From Table 3, we can infer that the parameters had mixed effects on the 3 consistencies taken under testing. The number of swallows were found to have increased as level of consistency increased. Average volume of swallow and speed of ingestion were found to have similar trend with highest values in level 0 consistency followed by levels 3 and 7. There was a drastic difference for total duration values between levels 0 and 3; however, minimal difference was noted on comparing duration values with level 3 and level 7. Hence, for the SPEAD parameter of total duration, no significant difference was found between level 3 and level 7.

The pairwise comparison results between the 3 stroke groups of varying severity are presented in Table 4.

Post hoc analysis from Table 4 portrays significant difference existed between minor and moderate groups and minor and moderately severe group. Participants with minor stroke had higher SPEAD rate as compared with moderate and moderately severe stroke, thereby indicating that participants with moderately severe stroke had poor performance on swallowing.

A 1-sample *t* test was conducted to compare the SPEAD rate in the stroke group against the normative value, which is summarized in Table 5.

Table 1 Results of Raw Data and ANOVA for SPEAD Test Across Stroke Severity

Consistency	Parameter	Stroke severity	Mean (SD)	Median (IQR)	*p* Value (ANOVA)
Thin liquids (level 0)	Number of swallows	Minor	6.4 (2.4)	6.5 (2.7)	0.068
		Moderate	8.6 (4.8)	7 (6)	
		Moderately severe	10.2 (4.8)	8 (5)	
	Total duration	Minor	25.1 (17.8)	17 (25)	0.022
		Moderate	49 (49.2)	32.5 (35.9)	
		Moderately severe	88.1 (60.9)	57 (51.4)	
Moderately thick liquids (level 3)	Number of swallows	Minor	10.6 (4.9)	9 (6.75)	0.876
		Moderate	9.9 (3.6)	10 (5)	
		Moderately severe	10.6 (2.6)	9 (2)	
	Total duration	Minor	61.7 (34.5)	53.8 (33)	0.043
		Moderate	71.6 (32.8)	67.2 (24.5)	
		Moderately severe	77.7 (8.2)	73.8 (7.9)	
Regular (level 7)	Number of bites	Minor	2.9 (2.1)	3 (2)	0.084
		Moderate	3.6 (2.2)	3 (1)	
		Moderately severe	3.6 (3.4)	2 (2)	
	Number of chews	Minor	63.9 (36.4)	53 (37.3)	0.389
		Moderate	65.1 (31.7)	61 (48.8)	
		Moderately severe	88.4 (36.7)	72 (29)	
	Number of swallows	Minor	3.2 (1.8.)	3 (2)	0.739
		Moderate	2.5 (1.5)	2 (1)	
		Moderately severe	2.8 (1.6)	2 (2)	
	Total duration	Minor	60.1 (35.8)	48.6 (29.1)	0.116
		Moderate	69.1 (64.1)	62.5 (37.5)	
		Moderately severe	94.5 (31.9)	103 (33)	

Abbreviation: SPEAD = Swallowing Proficiency for Eating and Drinking.

Table 2 Inter-Rater and Test-Retest Reliability Values

	Intraclass correlation			Cronbach alpha		
	Level 0	Level 3	Level 7	Level 0	Level 3	Level 7
Total duration	0.93	0.81	0.89	0.96	0.95	0.91
Number of bites	NA	NA	0.92	NA	NA	0.96
Number of swallows	0.92	0.97	0.96	0.93	0.96	0.93
Number of chews	NA	NA	0.90	NA	NA	0.81
Speed of ingestion	0.90	0.88	0.89	0.90	0.91	0.90
Average volume of swallow	0.92	0.91	0.91	0.84	0.86	0.84
SPEAD rate	0.90			0.91		

Abbreviations: NA = not applicable; SPEAD = Swallowing Proficiency for Eating and Drinking.

Table 3 Pairwise Comparison of SPEAD Indices Across Different Consistencies

Consistency		Mean difference	Standard error	*p* Value
Number of swallows				
Level 0	Level 3	−2.33	0.604	<0.001
	Level 7	−5.22	0.604	<0.001
Level 3	Level 7	−7.55	0.604	<0.001
Average volume of swallow				
Level 0	Level 3	4.31	0.815	<0.001
	Level 7	−12.25	0.815	<0.001
Level 3	Level 7	−7.94	0.815	<0.001
Total duration				
Level 0	Level 3	−24.62	10.2	0.005
	Level 7	24.88	10.2	0.004
Level 3	Level 7	0.62	10.2	1
Speed of ingestion				
Level 0	Level 3	2.74	0.337	<0.001
	Level 7	−4.43	0.337	<0.001
Level 3	Level 7	−1.68	0.337	<0.001

Abbreviation: SPEAD = Swallowing Proficiency for Eating and Drinking.

The results from Table 5 showed that the mean SPEAD rate in the stroke group was significantly lower than the normative value across all the age bands.

Pearson correlation was performed to estimate the strength of correlation between severity of stroke, SPEAD rate, and EAT-10 scores. The results are summarized in Table 6.

Table 6 presents negative moderate correlation between EAT-10 and SPEAD rate as well as between stroke severity and SPEAD rate, thereby indicating that increment in stroke severity and higher self-perception scores for swallowing impairment led to poor scores in the SPEAD test. A positive moderate correlation was observed between EAT-10 scores and stroke severity, indicating that participants with greater stroke severity reported a higher perceived difficulty with swallowing.

Discussion

The analysis of raw data, including the number of bites, swallows, chews, and total duration, revealed that these metrics were highest among participants with moderately severe strokes compared with those with less severe strokes. This finding may be attributed to disturbances in coordination and movement of tongue muscles associated with damage to the left hemispheric, suggesting that the oral phase of swallowing is predominantly controlled by the left hemisphere.[21] A study by Schimmel et al.[22] reported that despite the bilateral innervation of masticatory muscles, individuals with stroke demonstrated a chewing efficiency that was only half that of healthy controls. This impairment was attributed to reduced lip activity, as well as dysfunctions of the cheek and tongue. Individuals affected by stroke generally had prolonged oral phases and required more chewing cycles to process food.[23] Furthermore, a 2-year observational study

Table 4 Pairwise Comparison of SPEAD Rate Across the 3 Stroke Groups of Varying Severity

Severity of stroke		Mean difference	Standard error	*p* Value
Minor	Moderate	0.766	0.278	0.022
	Moderately severe	1.683	0.564	0.012
Moderate	Moderately severe	0.917	0.562	0.275

Abbreviation: SPEAD = Swallowing Proficiency for Eating and Drinking.

Table 5 Comparison of SPEAD Rate in Stroke Individuals With Healthy Normal Individuals

Age range	Statistic	df	*p* Value
21–40 y	−6.41	6	<0.001
41–60 y	−6.52	28	<0.001
>61 y	−7.83	40	<0.001

Abbreviation: SPEAD = Swallowing Proficiency for Eating and Drinking.
Note: The column labeled "Statistic" represents the t-value reflecting the difference between the sample mean and the corresponding population normative mean.

revealed that participants with stroke exhibited reduced masticatory ability and lower maximum lip restraining forces compared with healthy controls. Notably, these impairments did not improve over the duration of the study.[24] In individuals affected by stroke, oral stereognosis—the ability to recognize and discriminate objects within the mouth—has been linked to masticatory function.[25] Impaired intraoral sensation was associated with compromised masticatory function and unpleasant food hoarding in participants with stroke.[26]

Significant findings were observed in the SPEAD parameters, particularly in the SPEAD rate, which decreased with increasing severity of stroke. The SPEAD rate is calculated as the average of ingestion speeds across the 3 consistencies. This reduction may be attributed to a delayed swallowing reflex, a common occurrence in stroke patients.[27] In addition, the variety of lesions may play a role, as swallowing is a complex physiologic process involving multiple brain regions. Damage to specific areas of the brain can disrupt the initiation of the swallowing reflex. Lesions contributing to prolonged oral transit times were identified in the left superior and middle frontal gyrus, basal ganglia, and a small region of the insula, consistent with findings reported in previous research.[28] This suggests that the frontal cortex, involved in complex cognitive functions such as planning, decision making, and execution, is linked to delays in the oral phase of swallowing.[29,30]

Finally, we examined the association between stroke severity, EAT-10 scores, and SPEAD rate. The analysis revealed that an increase in stroke severity and a higher perception of self-reported swallowing difficulty were associated with a reduction in SPEAD rate. Central pattern generator (CPGs) in the brainstem are primarily reported to be involved in regulating swallowing.[1] Furthermore, any stroke-related damage to these CPGs could lead to reduction in SPEAD rate. In addition, cortical plasticity changes poststroke might lead to maladaptive swallowing behaviors, further affecting efficiency.[31] Other contributing factors include reduced sensory feedback from the oropharyngeal region,[32] muscle weakness or atrophy over time,[33] and psychological factors such as anxiety or fear of choking, which are common in individuals experiencing dysphagia after a stroke.[34]

Table 6 Pearson Correlation Matrix Across Self-Reported Swallowing Scores, Swallowing Function, and Severity of Stroke

	EAT-10	SPEAD rate	Stroke severity
EAT-10	—	−0.502	0.456
SPEAD rate	−0.502	—	−0.447
Stroke severity	0.456	−0.447	—

Abbreviation: SPEAD = Swallowing Proficiency for Eating and Drinking.

This study exhibited good to excellent test-retest reliability across the majority of parameters, except for the average volume of swallow across all consistencies, aligning with observations from previous research.[16,17] Similarly, interrater reliability yielded good to excellent scores based on evaluations by 2 speech-language pathologists, aligning with the results of the original article introducing the SPEAD protocol.[17]

Comparing the choice of Level 3 consistency used in the study by Dodderi et al.,[17] we chose to go ahead with pre packed curd of a specific brand (Milky Mist) to overcome the challenges faced by the participants in the abovementioned Indian study. The level 3 food item in our study was taken based on a thesis where various commercially available Indian food items were classified based on IDDSI guidelines and tested out across healthy individuals.[35] However, 3 individuals were excluded from the study due to history of lactose intolerance, and hence, their entire data were not considered for analysis.

Limitations include that the use of the SPEAD protocol, while effective, may not capture the full complexity of dysphagia in poststroke individuals, particularly when compared with other comprehensive swallowing assessments. Second, our study had less sample size which can affect the generalization of our findings. Future research should target larger sample size of similar cohorts and should consider confounding variables such as age and duration of dysphagia. In addition, the study relied on cross-sectional data, limiting the ability to evaluate changes in swallowing function over time or assess long-term recovery. Finally, external factors such as comorbidities, medications, and overall physical health were not accounted for during the analysis, which could influence the findings. Future studies should consider these factors and incorporate longitudinal designs to provide a more comprehensive understanding of dysphagia recovery in stroke survivors. Furthermore, future research can also focus on including larger cohorts and assess utility of SPEAD in individuals with nonoral modes of feeding as well.

This study evaluated swallowing function in poststroke individuals using the SPEAD protocol and identified a significant correlation between stroke severity and SPEAD

TAKE-HOME POINTS

- The Swallowing Proficiency for Eating and Drinking (SPEAD) protocol was able to capture measurable differences in swallowing efficiency across varying stroke severities, indicating its potential utility in clinical assessment of dysphagia in stroke survivors.
- An increase in stroke severity corresponded with prolonged ingestion durations and reduced ingestion speed, especially noticeable in moderately thick and solid consistencies.
- Integration of raw and derived parameters from SPEAD as well as self-reported swallowing difficulties (EAT-10) and neurologic assessment (NIHSS) offers a comprehensive understanding of dysphagia in poststroke population.
- A moderate correlation between SPEAD scores and EAT-10 emphasizes the value of combining clinician-assessed and patient-reported outcomes.

performance, with higher NIHSS scores associated with poorer outcomes. In addition, a reduced SPEAD rate and average volume of swallow was observed with increasing stroke severity. These findings emphasize the importance of quantifying swallowing impairments in stroke survivors. Furthermore, our findings suggest that SPEAD findings are not only feasible and informative in the acute and subacute settings but also align with the participants own perceptions of their swallowing abilities, as reflected in the EAT-10 scores. The moderate correlation between these 2 underscores the importance of incorporating both the clinician administered and patient-reported measures in dysphagia management. This dual approach will enhance clinical decision-making and promote patient-centered care.

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Author Contributions
U. Chilwan: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. A. Suresh: major role in the acquisition of data; analysis or interpretation of data. S. Shameer: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A. Zohara: major role in the acquisition of data. R. Balasubramanium: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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